PG – 300

III Semester M.C.A. Examination, April/May 2022
(CBCS – Y2K20 Scheme)
COMPUTER SCIENCE
3MCAE2 : Big Data Analytics (Elective)

Time : 3 Hours Max. Marks : 70

Instructions : *1) Part – **A** : Answer **any five** questions. **Each** carries **6** marks.*
*2) Part – **B** : Answer **any four** questions. **Each** carries **10** marks.*

PART – A

1. What is Big data ? What are the characteristics of big data ?
2. What is NOSQL ? Differentiate NOSQL from SQL.
3. What is data pre-processing ? What is the importance of data pre-processing ?
4. Write a note on data reduction.
5. Explain Bayesian classification in detail.
6. Explain spatial data mining.
7. Explain the ecosystem of HADOOP.
8. Explain Hive data types.

PART – B

9. Discuss the significance of big data in various industrial applications.
10. Explain data transformation technique in data pre-processing.
11. What is MapReduce ? Explain various levels in MapReduce workflow.
12. Explain HDFS in detail with a diagram.
13. What is Apache Pig ? Explain the various components of the model.
14. Write a note on following :
    a) Hbase **5**
    b) PigLatin. **5**

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JP – 772

III Semester M.C.A./M.Sc. (Two Years Course) Examination, May/June 2023

(CBCS – 2021 – 22)

COMPUTER SCIENCE

MCA3E2/MSC3E2 : Big Data and Analytics (Elective)

Time : 3 Hours Max. Marks : 70

Instructions : *1) Answer **all** Parts.*

*2) Part – **A** : Answer **any five** questions. **Each** carries **6** marks.*

*3) Part – **B** : Answer **any four** questions. **Each** carries **10** marks.*

PART – A **(5×6=30)**

1. Explain Big Data and its characteristics.
2. Explain the Hadoop ecosystem in detail.
3. Mention and explain any 2 data reduction techniques.
4. Explain classification using decision trees with an example.
5. Differentiate HBase and RDBMS.
6. What is Hive ? Explain its features and write brief notes on Hive Meta Store.
7. What is Pig ? Discuss its data processing operators.
8. How do you measure the prediction classification accuracy ?

P.T.O.

PART – B (4×10=40)

9. Write in detail the architecture of HDFS. 10

10. Write notes on :

    a) Data cleaning 5

    b) Data integration. 5

11. Write in detail the industrial applications of Big Data. 10

12. Explain the working of MapReduce. 10

13. What is clustering ? Explain various clustering techniques. 10

14. Write short notes on :

    a) Big SQL. 5

    b) NOSQL Vs RDBMS. 5

PG – 375

III Semester M.C.A. (Two Years Course) Examination, May 2024
(CBCS) (2021 – 22)
COMPUTER SCIENCE
MCA3E2/MSC3E2 : Big Data and Analytics (Elective)

Time : 3 Hours Max. Marks : 70

SECTION – A

Answer **any five** of the following. **Each** question carries **six** marks. **(5×6=30)**

1. List and explain any two industries that utilize Big Data.
2. Mention and explain any two data reduction techniques.
3. What is Hive ? Explain its features and write brief notes on Hive Meta Store.
4. What is Pig ? Discuss its Data Processing Operations.
5. Differentiate between supervised learning and unsupervised learning.
6. Explain any two techniques in Text Mining.
7. Explain Bayesian Classification in detail.
8. List any 3 big data challenges and explain.

SECTION – B

Answer **any four** of the following. **Each** question carries **10** marks. **(4×10=40)**

9. Describe with a diagram the different steps in Data Mining Process.
10. Explain in detail the architecture of a HDFS system.
11. Describe the two methods of handling missing values are addressed in a Dataset.
12. What is clustering ? Explain various clustering techniques.
13. Write short notes on the following :
    a) Hbase b) No SQL Vs. RDBMS.
14. State how to measure the prediction classification accuracy ? What are the metrics ?

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PG – 732

III Semester M.C.A. (Two Years Course) Examination, May/June 2025
(CBCS) (2021 – 22)
COMPUTER SCIENCE
MCA3E2/MSC3E2 : Big Data and Analytics (Elective)

Time : 3 Hours Max. Marks : 70

Instruction : *Answer **both** Sections.*

SECTION – A

Answer **any 5** questions. **Each** question carries **6** marks. **(5×6=30)**

1. What are the characteristics of big data and how do they differ from traditional data ?
2. Why is data pre-processing necessary ? Explain different data cleaning techniques.
3. Explain concept of Bayesian classification.
4. What is data mining ? With neat diagram explain steps of data mining.
5. Explain the process of partitioning and combining in MapReduce.
6. What is text mining ? Explain the process of text mining with diagram.
7. What is Hive Metastore ? Explain the differences between Hive QL and traditional SQL.

SECTION – B

Answer **any 4** questions. **Each** question carries **10** marks. **(4×10=40)**

8. Explain sources of big data. How do industries utilize Big data ?
9. Discuss the process of building decision tree for classification.

P.T.O.

10. Explain in detail architecture of HDFS.

11. What is data reduction ? Explain data cube and dimensionality reduction concept.

12. What are the various types of clustering algorithms ? Explain.

13. Explain the difference between HBase and traditional database. Explain how HBase store and retrieve data.

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PG – 674

III Semester M.C.A. (Two Years Course) Examination, May/June 2026
(CBCS) (2021-22)
COMPUTER SCIENCE
MCA3E2 : Big Data and Analytics (Elective)

Time : 3 Hours Max. Marks : 70

Instructions *: 1) Part – **A** : Answer **any five** questions. **Each** carries **6** marks.*
*2) Part – **B** : Answer **any four** questions. **Each** carries **10** marks.*

PART – A

1. Explain Big Data characteristics, sources and applications.
2. Explain and compare structured, semi-structured and unstructured big data types.
3. What is Data Preprocessing ? Why is Data preprocessing necessary ?
4. Given the following dataset showing marks of the students.

Student	Marks
A	40
B	60
C	75
D	80
E	65

Answer the following :

a) What is data normalization ? What is the advantage of normalization as a preprocessing technique ?
b) Apply Min-Max Normalization to scale the marks in the range [0,1].
c) Apply Z-score Normalization (Assume mean = 70 and standard deviation = 20).

5. Explain the anatomy of the File operations in HDFS.
6. Explain with illustrations any six PIG LATIN commands.
7. Compare HBase vs RDBMS in detail.
8. What is Collaborative filtering ? State any two applications of collaborative filtering.

P.T.O.

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PART – B

9. a) Explain the characteristics of big data. State the big data challenges. 4

   b) Explain any two Non-parametric techniques of Data Reduction. 6

10. a) Write a note on Data discretization and Concept hierarchy. 4

    b) Consider the following training set. Using Bayesian classification algorithm find the decision for the test data : (Sunny, Cool, Normal) 6

Instances	A1	A2	A3	Decision
1	Sunny	Hot	High	No
2	Sunny	Hot	High	No
3	Rainy	Hot	High	Yes
4	Rainy	Cool	Normal	Yes
5	Rainy	Cool	Normal	Yes
6	Sunny	Cool	High	No
7	Sunny	Hot	High	No
8	Sunny	Hot	Normal	Yes
9	Rainy	Cool	Normal	Yes
10	Rainy	Cool	High	Yes

11. a) Explain the MapReduce processing. Explain Map and Reduce functions with reference to the Wordcount MapReduce program. 8

    b) State any four basic Hadoop commands. 2

12. a) Discuss Hadoop ecosystem and its components. 6

    b) Explain with an example Java interfaces to HDFS. 4

13. a) What is HIVE ? State and illustrate any five HIVE QL commands. 6

    b) Write a note on Analytics with R. 4

14. Write short notes on :

    a) Text Mining. 5

    b) Clustering. 5

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